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MLS 101: Introduction to Medical Laboratory Science - Specimens, Safety and Accuracy

How laboratory results are produced, and why pre-analytical errors matter

FACULTY PRIMER

Faculty Primer

Faculty of Medical Laboratory Science
Medical Laboratory Science | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Most laboratory errors happen before the sample reaches the analyser. This pack introduces the laboratory disciplines, specimen collection and handling, biosafety, quality control and the meaning of reference ranges, with practical checklists for the teaching laboratory.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe the major laboratory disciplines and the specimen types each handles.
- Explain the pre-analytical, analytical and post-analytical phases of testing.
- Apply biosafety and waste-disposal rules in a teaching laboratory.
- Interpret quality-control results and explain reference ranges.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The laboratory disciplines

Medical laboratory science is a set of specialised diagnostic disciplines, each with its own methods and quality concerns.

- Haematology: blood counts, haemoglobin, coagulation and blood film morphology.
- Clinical chemistry: electrolytes, enzymes, glucose, renal and liver function.
- Medical microbiology: culture, identification and antimicrobial susceptibility testing.
- Histopathology and cytopathology: tissue and cell examination.
- Immunology and serology: antibody and antigen detection.

- Blood transfusion science: grouping, cross-matching and component therapy.

The three phases of testing

A result is only as reliable as the weakest phase in the chain.

- Pre-analytical: patient preparation, collection timing, correct tube, adequate volume, labelling, transport and storage.
- Analytical: calibration, reagent quality, instrument maintenance and technician competence.
- Post-analytical: result validation, reporting, critical-value communication and record retention.
- Haemolysis, delayed transport and under-filled tubes are common causes of rejected samples.
- Label at the bedside, never in advance, and never relabel a specimen.
- Critical results must be phoned to the clinician and documented.

> **Rejection is a safety feature**
> Rejecting a wrongly labelled sample protects the patient from a result belonging to someone else. Never 'fix' a label.

Biosafety and quality control

The laboratory is a controlled environment with rules that exist for a reason.

- Treat all specimens as potentially infectious.
- No eating, drinking, applying cosmetics or storing food in the laboratory.
- Use a biological safety cabinet for procedures that generate aerosols.
- Decontaminate surfaces before and after work; separate clean from dirty zones.
- Sharps into puncture-resistant containers; chemical and biological waste segregated.
- Quality control: run control materials, plot Levey-Jennings charts and stop testing when results fall outside acceptable limits.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Name the three phases of laboratory testing.

Answer: Pre-analytical, analytical and post-analytical.

2. Give three causes of specimen rejection.

Answer: Mislabelled specimen, inadequate volume, haemolysed sample, wrong container or delayed transport.

3. What is a reference range?

Answer: The interval within which results from a defined healthy population usually fall, used as a guide for interpretation and not a diagnosis.

4. Why are control materials run with patient samples?

Answer: To verify that the analytical system is performing within acceptable limits before patient results are reported.

5. State two biosafety rules for the laboratory.

Answer: Treat all specimens as infectious and never eat, drink or store food in the laboratory. Others include using PPE and decontaminating surfaces.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Match each laboratory discipline to two tests it performs.
- [ ] Write the three phases of testing and list two errors in each.
- [ ] Practise correct labelling technique with a supervisor.
- [ ] Demonstrate hand hygiene and PPE use in the laboratory.
- [ ] Interpret a simple Levey-Jennings chart and state when to stop testing.
- [ ] Learn ten laboratory safety rules and explain the reason for each.
- [ ] Complete one laboratory practical report with methods, observations and sources of error.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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